

How can stuff like ChatGPT 'consume' water if it just rains back down?

Posted by u/Total_Mountain_200 • 0 points • 5 comments

I always see stuff about how certain things use up all the water. Specifically, I saw that it takes 70 gallons of water to make 1 gallon of Coca-Cola, as well as all the stuff about ChatGPT using water.

But how do you 'consume' water? Doesn't it just rain back down? I see all this stuff like 'ChatGPT is draining the water supply!' but won't it just come back through the water cycle like a week later?

And how does it take 70 liters of water to make 1L of coke? 1L of coke comes in a 1L bottle, which means it can only hold 1L of liquid (duh). Where does the other 69L of water go? I assume it is used to absorb heat/other energy released during the process. But doesn't it just rain back down? How is it being 'consumed'?

Comments

u/Jusiena • 1 points • 2025-01-11 09:58 UTC

Water's on a nature-sponsored endless world tour, not consumed.

u/Wild_Thing_Nature • 2 points • 2025-01-11 09:48 UTC

Water is needed for essentially everything. I wouldn't be able to give you the specific breakdowns for your example, but for instance it would take water to run the machinery that pumps out the petroleum as well as to refine it into the plastic that then needs to be shaped into the bottle that holds the coke. (Not to mention the human factors that also need to consume water to survive while running or maintaining said machinery.)

In terms of your question about the water cycle - that is actually a very simplified way of looking at how water works in the environment. Yes, all water consumed will eventually make it back into circulation one way or another - but that doesn't mean it will end up in the same place it was taken from. For example, relatively near where I live are numerous underwater aquifers that took hundreds of years to form and fill, and which have relatively recently been drained over the last several decades much more quickly than it would take for them to replenish their supply (even without the recent ongoing droughts compounding the problem) - and now structures that were built above them such as homes and roadways have been collapsing into sinkholes because there are essentially big hollow-ish areas below them that used to be saturated with water.

Recall that I mentioned the human factor earlier as well - keep in mind that despite how much water there is on this planet, less than 1% is actually available for drinking water - shared amongst the 8 billion people that there currently are. Again, I don't know the logistics in terms of running machinery and such - but I know they wouldn't be using salt water for that, so that same water likely going to be used for all of the processes involved in production, and it needs to be cleaned before going back into the environment (or unfortunately in many cases, simply adds to the pollution in the environment). This is another way that water is "consumed."

u/Delehal • 4 points • 2025-01-11 09:37 UTC

On the one hand, yes, water is in some ways a renewable resource because it does cycle back around.

On the other hand, that cycle is not instantaneous. The vast majority of the water on Earth is not in a form that we can easily use. The amount of water that we can use is limited, and refills only so fast, and in many cases we use resources to treat it and make it safe to use.

If we drain aquifers, rivers, and lakes faster than they refill, they will run dry and that can devastate both local communities and local ecosystems. If we use water faster than it can be treated, that will exhaust the supply of safe drinking water for the surrounding area.

u/baes__theorem • 3 points • 2025-01-11 09:33 UTC

"consume" in this context means uses up, from a whole-process perspective.

coke doesn't have water as the only ingredient – it has sugar and other things in it, which all require water to be produced. the main water-consuming contributors are sugar and the manufacturing of the plastic bottle.

for large high-performance computing facilities – like those used by chatgpt and other llms – a lot of water is used to cool the servers.

when water is used up, it also doesn't simply "rain back down" as usable freshwater in a reservoir or something. lots of rain falls over the ocean, gets absorbed by the ground where it falls, or is otherwise non-capturable as a resource.

u/shootYrTv • 8 points • 2025-01-11 09:22 UTC

In places like California (where Silicon Valley is and therefore where most AI server farms are) where it rarely rains and these servers tap into the ground water or aqueducts, they are effectively consuming that water, because it will not rain down on them or feed back into their aqueducts or ground water. It will likely rain down into the ocean and become unusable saltwater